
Motioning: Selecting Movement–Sound Mappings Using Learning Progress

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Abstract

1 We present Motioning, a handheld interactive sound work that changes its
2 movement–sound mapping during use. An inertial measurement unit (IMU) inside
3 the interface estimates ten movement qualities. The system changes which of
4 these qualities affect a wind-like sound synthesized from white noise. An online
5 forward model predicts the participant’s movement one second ahead from the
6 current movement and mapping and is updated as new data arrive. At each mapping
7 change, the policy prioritizes qualities for which prediction error is decreasing
8 and suppresses qualities that have recently dominated the interaction. This se-
9 lection is designed to interrupt convergence on a fixed movement–sound relation
10 and reopen the search for a next movement. The model selects the qualities to
11 which the sound responds but does not determine the participant’s movement. We
12 describe the system and selection policy, verify the adaptive loop in studio tests,
13 and report observations from an earlier prototype exhibition that did not use the
14 learned policy.

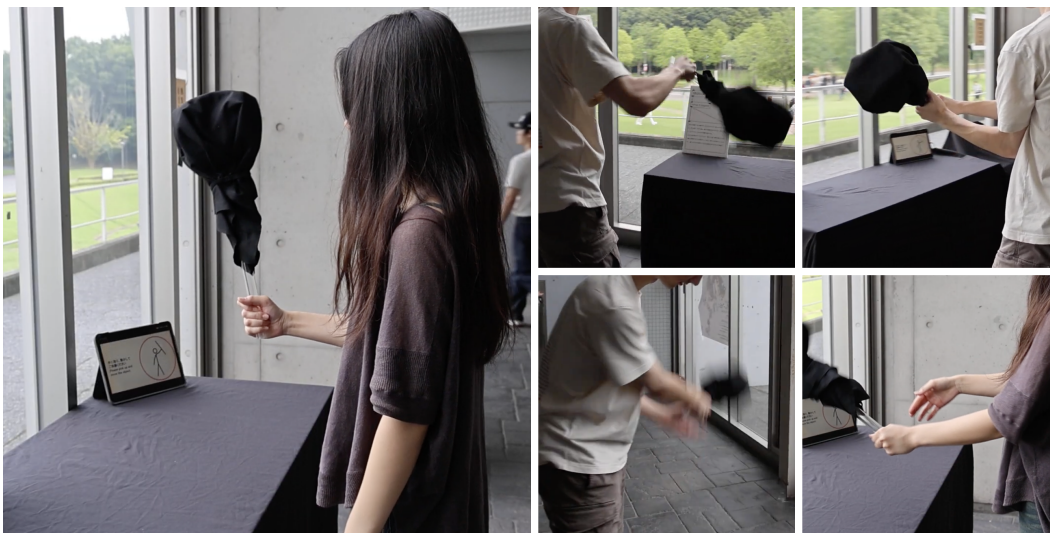


Figure 1: Five views of visitors moving the rod-shaped interface at the July 2026 university exhibition. This prototype used a rule-based mapping policy and did not include the learned policy. Video documentation is available at https://youtu.be/g05JIs_c2JA.

1 Introduction

Movement–sound mapping specifies the relation between input at a performance interface and parameters of a sound process. A clear and consistent relation is often quick to learn. A mapping that connects several inputs and outputs is harder to exhaust than a one-to-one relation and can leave room for a performer to try different actions [8]. Motioning uses this difficulty of fully mastering a mapping as a condition for movement exploration.

Prior work indicates that sound–gesture mappings affect perceived action possibilities [1] and that movement sonification may expand a person’s movement repertoire [10]. Here, *movement exploration* means trying trajectories, postures, speeds, and timings not previously used in the interaction, guided by changes in the sound’s response.

In Motioning (Figure 1), a participant moves a rod-shaped interface while the system changes the qualities to which the sound responds. A wind-like sound changes with estimated qualities such as balance, rotation, acceleration, and stillness. The system foregrounds two of ten qualities. It selects the next pair using learning progress: the recent decrease in the online forward model’s prediction error [11]. High error alone does not raise priority. The policy also suppresses qualities that have recently been prominent in the movement or selected for the mapping. It therefore favors relations that the model is learning but that have not yet dominated the interaction.

A quality can be produced through several trajectories, postures, speeds, and timings. The artist defines the quality vocabulary, sound, and selection rule; the model selects a mapping within those constraints; the participant selects the concrete movement. Motioning treats improvement in the model’s prediction as a proxy for the formation of a movement–sound expectation. Changing the mapping then alters the conditions for the participant’s next movement.

2 Related Work

2.1 Movement–sound mappings for exploration

In an electronic instrument, the mapping between an input device and a sound source affects both the character of the instrument and the performer’s response [8]. Direct and consistent relations support control. Relations whose outcome cannot be grasped at once give a performer a reason to try different actions. Motioning changes the sound’s response target after partial mastery, making the relation available for exploration again.

Sound–gesture relations can provide cues for a next movement. Altavilla et al. found that sound–gesture mappings contributed to the gestural affordances participants perceived in sound [1]. Franić’s *Flo)(ps* couples habitual and unusual actions with different sonic responses, using sound to draw attention beyond familiar ways of handling an everyday object [7]. Oppici et al. propose sonification that broadens the set of movements a person explores [10]. Motioning changes the response target so that a participant repeating one operation must search for the bodily actions to which the sound now responds.

Interactive machine learning and live coding allow movement–sound relations to be changed during use. Wekinator lets a designer iteratively demonstrate, train, and evaluate supervised mappings in real time [3]. Motion-Sound Mapping through Interaction learns mappings from users’ corporeal demonstrations [4]. Scurto et al. use online clustering to let a user shape a motion–sound mapping [12], while Co-Explorer pairs a user with a reinforcement learning agent that explores a sound-parameter space [13]. In CO/DA, a live coder rewrites movement–sound relations during dance improvisation [5]. These systems change a relation through demonstration, parameter exploration, or human intervention. Motioning uses an online model to rank artist-defined candidates and change which sensed movement qualities become sonically salient.

2.2 A vocabulary for movements without gesture names

Motioning estimates qualities such as balance, rotation, acceleration, and tension rather than recognizing gesture names. A movement that does not match an operation such as “swing” or “rotate” can still enter the system as a combination of qualities.

64 Miyoshi’s *kinaesthetic elements* describe the sensations that movements of non-human objects evoke
65 in an observer through a vocabulary of fifteen fragments [9]. Motioning does not use this vocabulary
66 to record movement sequences or return named gesture classes. It uses the elements as dimensions
67 for continuous estimation, separating object movement into qualities such as balance, rotation,
68 articulation, and rhythm.

69 The Effort category of Laban Movement Analysis also describes qualitative dimensions of movement
70 and has been used for recognition from multimodal body-sensing data [2] and sonification [6].
71 Effort primarily describes how a mover exerts energy. Kinaesthetic elements describe the movement
72 sensation that a moving object evokes in an observer. In Motioning, the participant moves the object
73 while perceiving its motion, so acting on the object and observing its movement occur within the
74 same loop.

75 2.3 Selecting mappings from learning progress

76 Motioning uses machine learning to select mappings. The model predicts the participant’s next
77 movement under the current mapping and measures improvement for each quality. Oudeyer et al. use
78 this learning progress to select exploratory situations while avoiding those that are already predictable
79 or too irregular to learn [11].

80 Motioning uses learning progress to select the mapping presented to a participant, rather than an
81 exploratory situation for an autonomous system. It prioritizes qualities whose prediction error is
82 decreasing and suppresses qualities that have recently been prominent or selected. Applying learning
83 progress in this way assumes that improved model prediction tracks an emerging movement–sound
84 expectation; this assumption requires evaluation.

85 3 Motioning

86 Motioning consists of a rod-shaped interface containing an IMU and a sound and learning system
87 running on a laptop. When the participant tilts, swings, rotates, or stops the rod, the IMU captures
88 linear acceleration and angular velocity; the system uses these data to estimate ten movement qualities.
89 The rod form may affect which movements participants try, but we have not yet compared it with
90 other interface forms.

91 3.1 Rod-shaped interface and movement qualities

92 The interface contains one six-axis IMU. Because the participant holds the interface rather than
93 attaching the sensor to the body, the body and object move together while the participant hears the
94 sound’s response to the object’s motion.

95 Miyoshi defines fifteen kinaesthetic elements [9]. Motioning implements the ten that can be estimated
96 from one IMU: balance, rotation, articulation, acceleration, deceleration, gravity, vibration, rhythm,
97 tension, and stillness. The remaining elements require information about the body, surrounding space,
98 or surface contact that the IMU does not provide. Tension is particularly indirect because the IMU
99 measures motion associated with elastic force rather than force itself. Appendix A summarizes the
100 estimates.

101 3.2 Interaction and sound

102 IMU data reach Max/MSP over OSC (Appendix C). Each feature is normalized online to the
103 participant’s movement range, so no separate calibration stage is required.

104 The sound is a wind-like material synthesized from white noise (Appendix B). The ten movement
105 qualities affect the envelope, filter, and granularity of the same material. A mapping change preserves
106 the material while changing the qualities to which it responds most strongly. The system always
107 selects two qualities and assigns a larger weight to one and a smaller weight to the other.

108 A continuous *match* value measures agreement between the current mapping and the estimated
109 movement qualities. It is the inner product $M_t = \mathbf{m}_t \cdot \mathbf{a}_t$ of the mapping weights and the normalized
110 activity vector. Higher activity in the selected qualities therefore increases match, which opens the
111 filter and raises the gain. This response provides a cue for finding the current target.

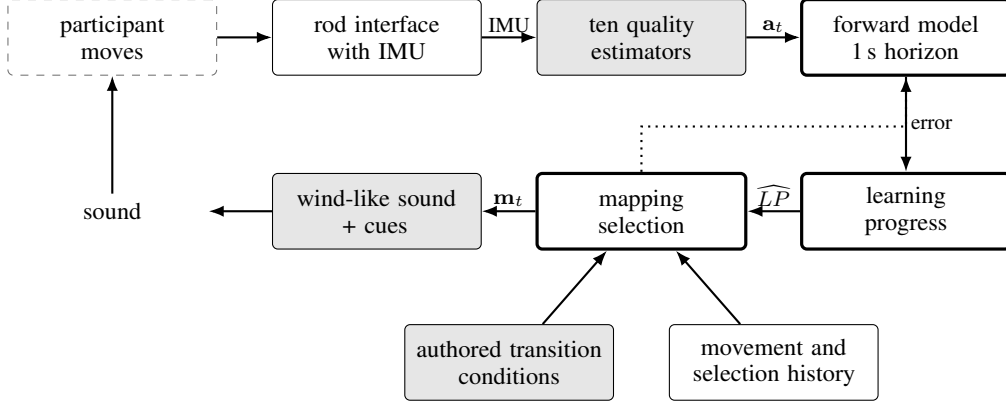


Figure 2: The closed interaction loop. Movement qualities are estimated from the rod’s IMU and change a wind-like sound according to the current mapping. The forward model predicts movement conditioned on that mapping. Mapping selection combines learning progress, movement history, selection history, and artist-defined transition conditions.

112 A sustained drone and a short mapping-change sound were added after the prototype exhibition in
 113 Section 3.4. When the participant was moving but match was low, the wind-like sound alone did not
 114 clearly indicate that the system was operating. The drone now sounds when movement is present and
 115 match is low, indicating that the system is still receiving movement data. A short sound marks each
 116 change in the response target.

117 3.3 Mapping policy

118 The online forward model updates whenever it receives new sensor data and predicts movement
 119 qualities one second ahead from the current movement and mapping. Its input comprises current
 120 activity $\mathbf{a}_t$, its first difference $\Delta \mathbf{a}_t$, the current mapping $\mathbf{m}_t$, and their element-wise product:

$$\hat{\mathbf{a}}_{t+1} = f_{\theta}(\mathbf{a}_t, \Delta \mathbf{a}_t, \mathbf{m}_t, \mathbf{a}_t \odot \mathbf{m}_t). \quad (1)$$

121 The linear model is updated at 20 Hz by recursive least squares and reset for each participant. It
 122 tracks local changes in predictability within a short interaction rather than modeling long movement
 123 sequences.

124 For each quality, prediction error E_i is the difference between the predicted movement and the
 125 movement observed one second later. Learning progress LP_i compares error over slow and fast time
 126 scales:

$$LP_i(t) = \max(0, E_i^{\text{slow}}(t) - E_i^{\text{fast}}(t)). \quad (2)$$

127 This assigns a positive value only while prediction is improving; irregular movement with persistently
 128 high error does not receive a high value. The fast and slow averages use 2- and 8-s time constants,
 129 and a decaying running maximum normalizes learning progress to $\widehat{LP}_i \in [0, 1]$.

130 Conceptually, the candidate score combines a baseline and learning progress with penalties for
 131 movement and selection history and a constraint on eligible transitions:

$$S_i \propto (\text{baseline} + \widehat{LP}_i) \times \text{movement-history penalty} \times \text{selection-history penalty} \times \text{transition condition}. \quad (3)$$

132 The baseline permits selection at zero learning progress; the history terms downweight recently
 133 prominent or selected qualities; and the transition condition is an artist-authored weight from the
 134 current primary quality to candidate i . Mappings change at intervals, but a high match retains the
 135 current mapping to avoid interrupting engagement with that relation.

136 With synthetic input, learning progress rose temporarily for a learnable periodic signal and did not
 137 rise for random or constant input. Studio trials confirmed that the same policy changed mappings in
 138 response to live movement.

139 3.4 Observations from a prototype exhibition

140 In July 2026, approximately thirty visitors used a prototype at an exhibition held within the university.
141 This version changed mappings through a rule-based policy and did not include the learned policy.
142 Some visitors put down the rod before finding a movement–sound relation. In several cases, sustained
143 movement began after a spoken prompt. These observations led to the drone and mapping-change
144 sound described in Section 3.2.

145 Comments were collected through a voluntary, gallery-wide feedback form with an open comment
146 field; no identifying information was collected. Some comments described discomfort with conspicu-
147 ous movement in public. One visitor said that they became “a strange person waving a stick around,”
148 and another asked for a booth in which to move alone. Other visitors said that the changing mapping
149 kept them interacting with the work, or contrasted attaching sound to movement with using sound to
150 change movement. In one unplanned encounter, a visitor swung the rod while the artist danced to the
151 resulting sound.

152 4 Discussion

153 4.1 Roles of mapping and object form in movement exploration

154 During the prototype exhibition, the artist observed a range of bodily movements that had not been
155 anticipated during the work’s design. Motioning selects a quality rather than a movement sequence.
156 Selecting rotation, for example, does not determine the participant’s trajectory, posture, speed, or
157 timing, so the same mapping can be realized through different movements.

158 Interface form constrains movement alongside the sound mapping. The rod form suggests gripping,
159 swinging, and rotating. In the encounter where a visitor moved the device and the artist danced,
160 sensor control and bodily response were split between two people. Future versions could compare
161 other object forms, including objects designed to be held by several people, and examine how form
162 changes both the movements and the number of participants in the loop.

163 Machine learning does not select the participant’s movement, but it steers attention by selecting the
164 qualities to which the sound responds. The artist limits the available sounds and transitions; the model
165 changes the response target; the participant decides how to move, whether to follow it, or whether to
166 stop. AI may support human agency by varying conditions for action without prescribing their exact
167 physical output.

168 4.2 Limitations and future directions

169 Studio trials verified that the learned policy operated during live interaction, but did not compare
170 movement with and without the learning-progress term. Such a comparison should measure movement
171 coverage and document how participants respond to, resist, or fail to notice mapping changes.

172 Improvement in the model’s prediction does not establish that the participant understands the mapping.
173 Prediction error may decrease when a participant repeats a movement without recognizing its relation
174 to sound. Recognition of the current response target therefore requires a separate measure. Also,
175 the installation setting may also affect participation: some visitors reported discomfort with moving
176 conspicuously in public. An evaluation should either control the setting or include it as a factor in the
177 analysis.

178 5 Conclusion

179 Motioning estimates movement qualities from a rod-shaped interface and changes the qualities to
180 which a wind-like sound responds during interaction. An online forward model predicts movement
181 under the current mapping. The system selects the next mapping by combining improvement in its
182 predictions with recent movement and selection history.

183 The changing response is intended to reopen the search for a next movement. Studio trials verified
184 the adaptive loop; whether learning-progress-based selection broadens exploration relative to fixed or
185 rule-based mapping remains to be tested.

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Table 1: The ten implemented elements. Definitions paraphrase Miyoshi [9]; estimates use a handheld inertial unit.

Element	Kinaesthetic sensation	Estimate from the inertial unit
Balance	Simulated sense of orientation, salient once equilibrium is lost	Deviation of the gravity vector from rest attitude
Rotation	Centrifugal force felt while moving along a curve	Gyroscopic self-rotation combined with an accelerometer estimate of travel around a more distant axis
Articulation	Movement of a skeletal structure within its degrees of freedom	A growing arc followed by cessation
Acceleration (<i>impulse</i>)	Positive inertia; transition from stasis to motion	Positive slope of a smoothed acceleration envelope
Deceleration (<i>resistance</i>)	Negative inertia; reduction in speed	Negative slope of the same envelope
Gravity	Awareness of vertical gravitational force	Vertical component of linear acceleration
Vibration	Rapid shaking perceived as an integrated whole	High-band acceleration energy from a Goertzel filter bank
Rhythm	Oscillation slow enough for each stroke to be felt separately	Autocorrelation over movement vectors: periodicity rather than energy
Tension	Felt elastic force of stretching or twisting	Sustained low-band acceleration energy
Stillness	The quality of being immobile	Low linear and angular energy jointly

A The ten kinaesthetic elements and their estimation

Miyoshi treats vibration and rhythm as a frequency continuum: rhythm is vibration slow enough for individual strokes to be felt separately. The implementation approximates this continuum only loosely. It estimates vibration and tension from high and low bands of the same acceleration-energy signal, placing tension on a frequency axis absent from Miyoshi’s definition. Rhythm is estimated from periodicity rather than energy and therefore rests on a different measure.

B Sound design

The sound is a wind-like material generated from filtered white noise. Each movement quality affects its envelope, filter trajectory, and granularity. As match rises, gain and filter opening rise together. A sustained drone indicates continued system operation when the participant is moving but match is low. A short contact sound marks a change in the primary quality.

C Software

The work runs on a laptop. The IMU inside the rod-shaped interface sends inertial data to Max/MSP over OSC. A JavaScript object estimates and normalizes the qualities, updates the forward model, and selects mappings; Max generates the sound. The same control code runs outside Max for tests with synthetic input.